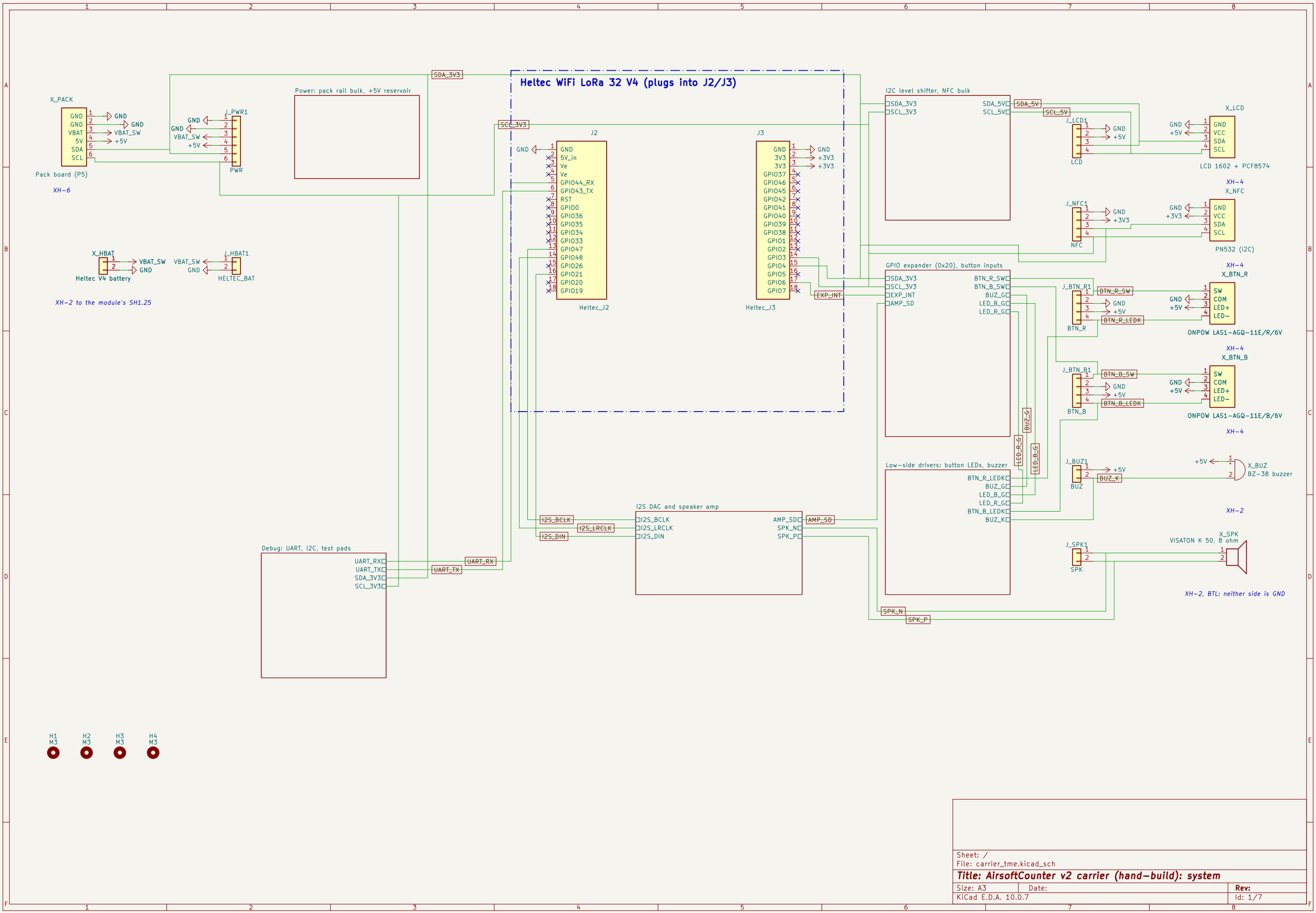
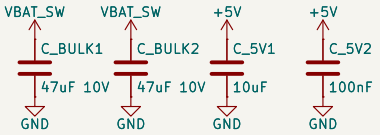

Heltec WiFi LoRa 32 V4 (plugs into J2/J3)
X_PACK
GND
GND
VBAT
5V
SDA
SCL
Pack board (P5)
XH-6
J_PWR1
PWR
Power: pack rail bulk, +5V reservoir
X_HBAT
VBAT_SW
GND
Heltec V4 battery
J_HBAT1
HELTEC_BAT
XH-2 to the module's SH1.25
SDA_3V3
SCL_3V3
J2
5V_in
Ve
GPIO44_RX
GPIO43_TX
RST
GPIO0
GPIO36
GPIO35
GPIO34
GPIO33
GPIO47
GPIO48
GPIO26
GPIO21
GPIO20
GPIO19
Heltec_J2
J3
3V3
GPIO37
GPIO46
GPIO45
GPIO42
GPIO41
GPIO40
GPIO39
GPIO38
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
EXP_INT
Heltec_J3
I2C level shifter, NFC bulk
SDA_5V
SCL_5V
J_LCD1
LCD
X_LCD
VCC
LCD 1602 + PCF8574
J_NFC1
NFC
X_NFC
PN532 (I2C)
GPIO expander (0x20), button inputs
AMP_SD
BTN_R_SW
BTN_B_SW
BUZ_G
LED_B_G
LED_R_G
J_BTN_R1
BTN_R
X_BTN_R
SW
COM
LED+
LED−
ONPOW LAS1-AGQ-11E/R/6V
J_BTN_B1
BTN_B
X_BTN_B
ONPOW LAS1-AGQ-11E/B/6V
Low-side drivers: button LEDs, buzzer
BTN_R_LEDK
BTN_B_LEDK
BUZ_K
J_BUZ1
BUZ
X_BUZ
BZ-38 buzzer
XH-2
J_SPK1
SPK
X_SPK
VISATON K 50, 8 ohm
XH-2, BTL: neither side is GND
I2S DAC and speaker amp
I2S_BCLK
I2S_LRCLK
I2S_DIN
SPK_N
SPK_NC
SPK_P
SPK_N
SPK_P
Debug: UART, I2C, test pads
UART_RX
UART_TX
H1
H2
H3
H4
M3
Sheet: /
File: carrier_tme.kicad_sch
Title: AirsoftCounter v2 carrier (hand-build): system
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.7
Id: 1/7

Power in (from pack board)

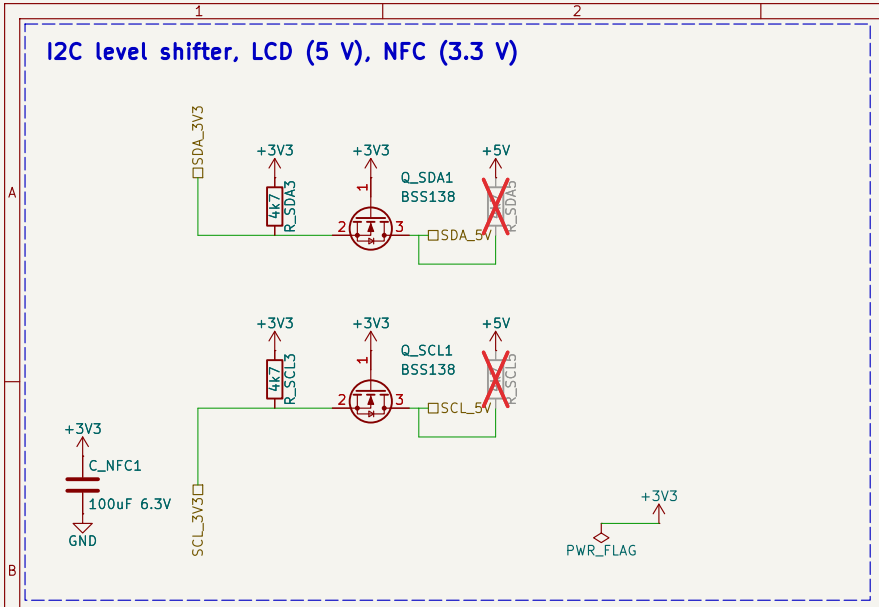




Sheet: /Power: pack rail bulk, +5V reservoir/
File: carrier_tme-power.kicad_sch

Title: AirsoftCounter v2 carrier (hand-build): Power: pack rail bulk, +5V rese

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 2/7

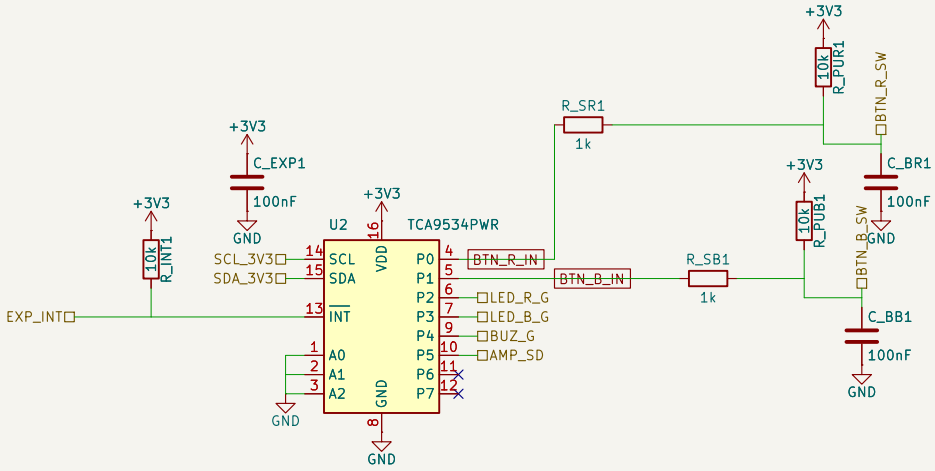


Sheet: /I2C level shifter, NFC bulk/
File: carrier_tme-i2c.kicad_sch

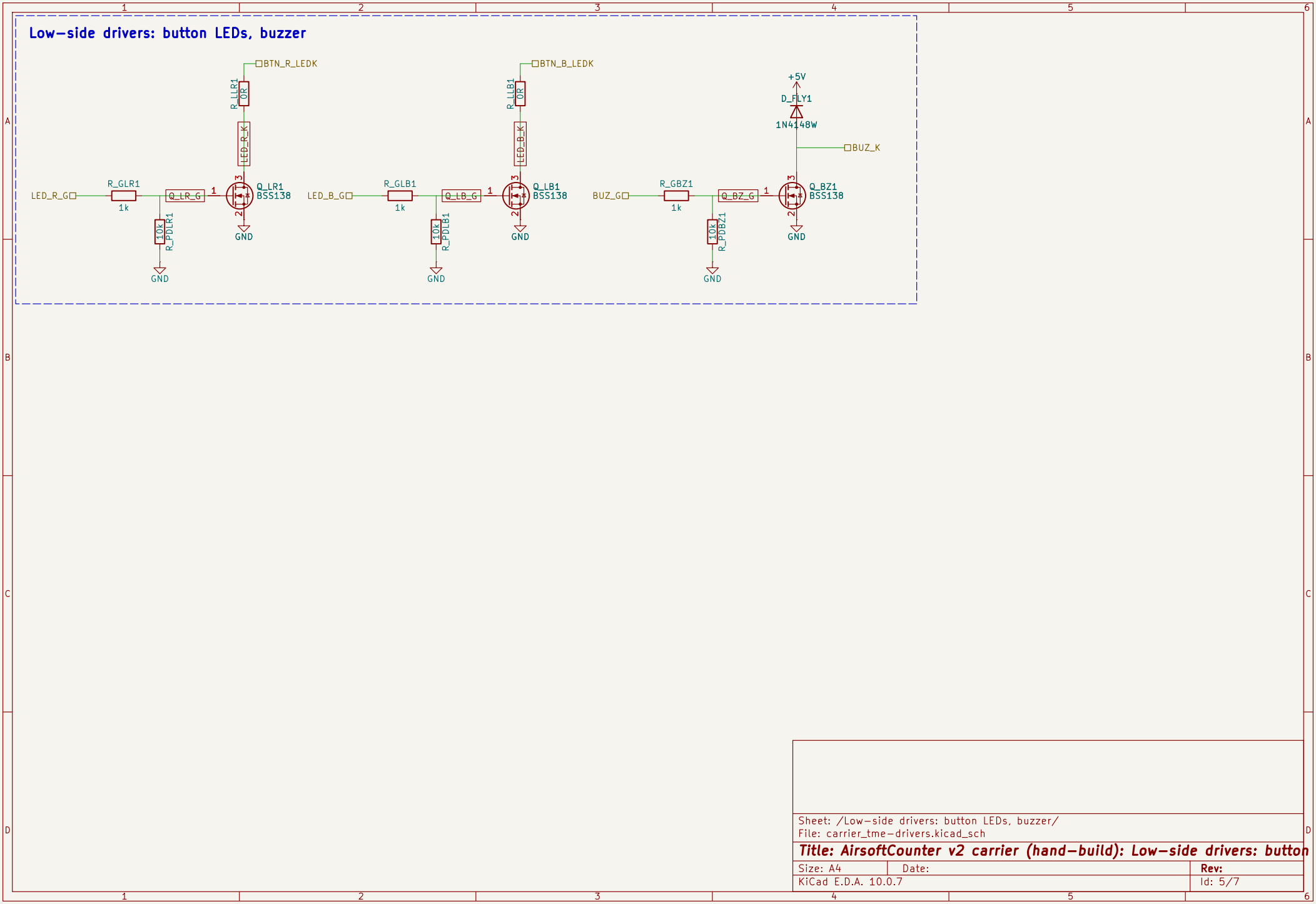
Title: AirsoftCounter v2 carrier (hand-build): I2C level shifter, NFC bulk

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.7		Id: 3/7

GPIO expander (0x20), button inputs



Sheet: /GPIO expander (0x20), button inputs/		
File: carrier_tme-expander.kicad_sch		
Title: AirsoftCounter v2 carrier (hand-build): GPIO expander (0x20), button i		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 4/7

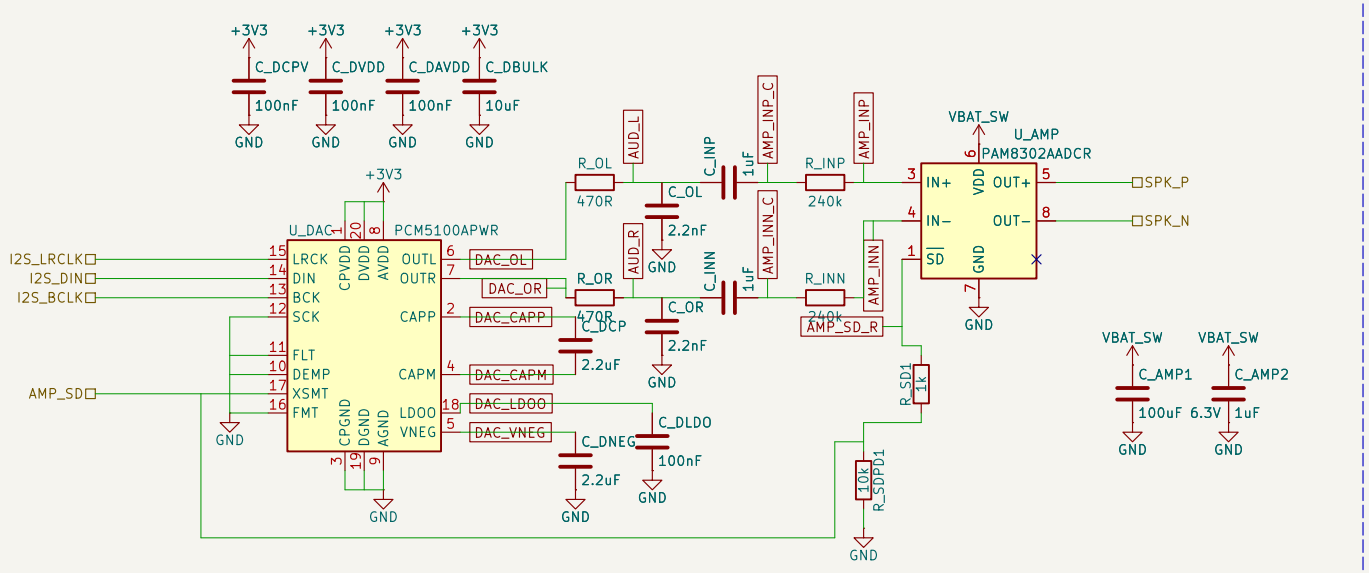


Sheet: /Low-side drivers: button LEDs, buzzer/
File: carrier_tme-drivers.kicad_sch

Title: AirsoftCounter v2 carrier (hand-build): Low-side drivers: button LEDs,

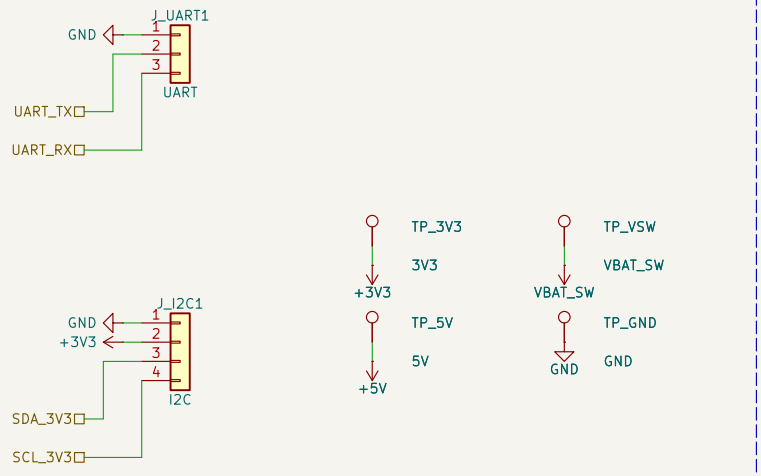
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.7		Id: 5/7

I2S DAC and speaker amp



Sheet: /I2S DAC and speaker amp/		
File: carrier_tme-audio.kicad_sch		
Title: AirsoftCounter v2 carrier (hand-build): I2S DAC and speaker amp		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 6/7

Debug: UART, I2C, test pads



Sheet: /Debug: UART, I2C, test pads/		
File: carrier_tme-debug.kicad_sch		
Title: AirsoftCounter v2 carrier (hand-build): Debug: UART, I2C, test pads		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 7/7